



The future of membrane gas separations

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ABSTRACT

Membrane processes are nowadays considered as a key technology for gas separation applications. Except the very specific use of mineral membranes multistaged process for uranium enrichment performed since 1942, it took more than 100 years before the visionary application of selective materials, proposed by T. Graham in 1866 for gas separation purpose, became an industrial reality. Major issues in terms of materials science, module manufacturing and process design methods needed indeed to be solved for the membrane solution to be considered as reliable and cost-effective compared to other gas separation technologies. Air separation, hydrogen purification, natural gas and biogas upgrading, VOC recovery and gas drying currently are the major industrial applications. The number of membranes and modules suppliers remains however very limited. Moreover, the commercially available membrane modules almost exclusively make use of ultra-thin dense polymers on a porous support, with a very limited choice in terms of material active layer. Taking into account the major recent developments in nanostructured materials, production technologies (such as 3D printing) and advanced process design methods, the possible, highly prospective evolutions in the field are discussed.

1. Introduction

Membranes are one of the major current gas separation technologies and offer reliable and cost-effective solutions for several industrial applications [1,2]. It took however a long time for the visionary concept of gas separations thanks to a selective material, proposed by Thomas Graham in 1866 [3], to become a reality. The first large scale membrane gas separation unit is indeed usually dated 1980 for hydrogen recovery purpose [4]. Major challenges had in fact to be addressed such as the production of defect free thin skin polymeric materials, robust hollow fibers and module manufacturing technologies, and effective process design methods [5]. These efforts required years of research and development and were produced by a few pioneering companies (Air Products, Dow, Separex, Cynara, Grace, Dupont, Ube) [4]. Looking at the current context of membrane gas separations in industry, it is interesting to notice that most of the polymeric materials families and the major industrial applications operated today were already identified and developed some four decades ago.

Given the major developments recently achieved in selective materials synthesis, fabrication methods and process systems engineering tools, it might be that significant changes will occur for membrane gas separations applications in the near future. The fast changing chemical process industries (CPI) landscape is also generating new separation challenges and new specifications.

This prospective analysis intends to explore the potential evolution in the field. Starting from the general framework and historical overview of membrane gas separations, the industrial membrane gas separation current situation is presented, and a series of future developments are discussed. The recent advances in nanostructured high performances membrane materials, module production technologies and process design methods are more specifically analyzed. It is expected that, based on an overview of the past and present of membrane gas separations, the analysis of the future changes of the technology and tentative evolution tracks can be (partly) foreseen.

2. Historical overview

The very first observation of a membrane separation effect is classically attributed to Abbé Nollet, who reported in 1752 the rupture of a pig bladder sample when put in contact with two different liquid solutions (water and an ethanol aqueous solution) in a closed vessel [6]. The boiling effect interpretation proposed by Nollet was wrong. The material deformation and breakage were later attributed to the mechanical forces exerted on the membrane, due to a forward osmosis phenomenon [7].

Some hundred years later, Thomas Graham explored for the first time the transport rate of different gases through natural rubber balloons [3]. The significant differences of the balloon collapse rate from one gas to another suggested a non-viscous flow phenomenon. A dense

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structure of natural rubber was thus postulated, with a strict diffusional mass transfer effect. A so-called solution-diffusion mechanism was then proposed to account for the different experimental permeability results. This visionary concept is still considered today as the cornerstone of gas transport modelling and interpretation into dense polymers [8]. Graham also showed that the gas mixture separation efficiency was not affected by the membrane thickness and postulated that membrane materials could possibly achieve gas separations for industrial purpose one day [3].

It took however one century for this idea to become reality, with the first large scale application of polymeric membranes for hydrogen recovery by Monsanto with Prism permeators in 1980 [5,9]. Tremendous research and development efforts were indeed required in order to achieve robust and efficient thin skin polymeric membranes production and defect free module fabrication technologies. It is important to note that these challenges have been addressed thanks to the cross-fertilization of technology developments from the development of reverse osmosis membranes and modules in the 1960s. Interestingly, some of the earliest gas membranes were developed by controlling the drying process of RO membranes and modifying the RO membrane formation process. In parallel, a dedicated process design methodology was also necessary, in order to adequately evaluate the separation performances and cost effectiveness of membranes when compared to older and mature technologies such as cryogeny or absorption. The process engineering knowledge developed in the 40's for the isotope separation challenge was of major interest for that purpose [10–12].

3. Membrane gas separations today

Membrane processes are nowadays included in the portfolio of mature industrial gas separation processes, together with cryogeny, absorption and adsorption operations (Table 1) [1,2]. It is important to note that the decision to develop membrane gas separation technology required at the time a significant paradigm shift and associated risk. From a conceptual point of view, membranes make use of an intrinsic rate separation effect, while the other technologies (thermal or auxiliary phase) are systematically based on a phase equilibrium effect [13]. Membranes can thus be considered as the third wave of separation processes [14,15].

The difficulties of ultrathin material production, the mechanical, thermal and chemical resistance issues, the technological problems of multi sheets or fibers module fabrication explain to a large extent why membranes lately entered the field. Moreover, the scale-up specificity of membrane technology, through a module numbering up solution, was completely different to the three other technologies, which benefit from unit large scale apparatus possibilities. This specificity is still often considered today as a drawback by industry decision makers and can lead to the early rejection of the membrane solution for large capacity projects. Hopefully, large scale applications with thousands of modules and million membrane square meters plants show that the membrane solution is not limited to small scale problems [4]. A numbering up

Table 1
Summary of the four major gas separation processes.

Separation Process	Separating agent	Process Type	Energy input	First industrial application
Cryogeny	Thermal	Equilibrium	Work (cooling)	~1900
Absorption	Auxiliary phase (liquid)	staged	Heat (chemical)	~1910
Adsorption	Auxiliary phase (solid)		Work (physical)	
			Heat (TSA) and/or work (PSA, VSA)	~1950
Membranes	Solid permselective material	Rate controlled	Work (compression/vacuum)	~1980

process does not necessarily imply a costly solution.

Besides the different drawbacks listed above, it is however important to point out the key advantages of the membrane gas separation solution: continuous operation (no complex flowsheet such as cyclic multi-column adsorption), no regeneration step (simple process and flowsheet compared to absorption and adsorption), no phase change (possibility of energy efficient processes), compact units due to the large specific surface area of modules (intensified process), physical separation process with no chemical reactions, no chemicals losses, no waste generation (environmental friendly process).

These specificities gradually opened the way for key applications covering a broad range of industrial sectors, summarized in Table 2: air separation, hydrogen purification, natural gas and biogas treatment, Volatile Organic Compounds (VOC's) recovery and gas drying. These five operations correspond to the major current markets of membrane gas separations. It is important to note that current applications are limited to ambient or moderate temperature conditions (typically 80C maximum), due to the limited thermal resistance of polymers and modules. The overall economical market is difficult to evaluate but is considered to be around 5 to 15 billions \$ per year, with a growth rate above 10 % [16]. Nitrogen Enriched Air (NEA) is by far the largest in terms of market share (ca 50 %) and number of units [17].

More recently, carbon capture has emerged as the most actively investigated topic of membrane gas separations, especially in terms of materials development. For post-combustion application, some industrial pilots and demonstrators show that membranes can compete to amine absorption, the currently best available technology. It might be that carbon capture will become a very significant market for membranes in the near future, in a sustainable and carbon neutral framework, for power plants (coal, gas), cement, steel and refineries emissions.

Promising gas or vapor applications are expected to possibly emerge in the near future. More generally, advanced membrane materials are expected to offer an energy efficient solution to large scale separations of major interest, such as discussed by Scholl and Lively [18].

It is interesting to note that most of the applications and membrane materials listed in Table 2 were identified and developed from the very beginning (NEA, H₂ recovery, natural gas upgrading installations started in the 80's [5]). The VOC application with, for the first time, an elastomeric active layer, started a decade later. Even though large improvements over the years in the separation performances of the membrane materials listed in Table 2 have to be acknowledged, it is striking that membranes have shown an impressive continuing development in terms of market thanks to a very limited set of materials and

Table 2

Overview of the major current gas separation applications of membranes (adapted from 4, 9, 17). Carbon capture is still an emerging application showing a very large market potential.

Application	Membrane material	Selectivity (α)	Module
Air separation	Polysulfone, polyimide, poly (phenyleneoxide)	4–7 O ₂ /N ₂	Hollow fiber
Hydrogen purification	Polysulfone, polyimide	~100 H ₂ /N ₂ H ₂ /CO	Hollow fiber
Natural gas and biogas upgrading	Cellulose acetate, polyimide	15–40 CO ₂ /CH ₄	Spiral or Hollow fiber
VOC recovery	Silicone rubber	10–30 VOC/N ₂	Spiral or Disc
Drying	Polyimide, polysulfone, Nafion	>200 H ₂ O/N ₂	Hollow fiber (Bore side feed)
Carbon capture (post-combustion)	Block copolymers (e.g. poly (ether-amide), poly(ethylene oxide) – poly(butylene terephthalate)	~50 CO ₂ /N ₂	Plate & frame, spiral

around five key applications. This matter of fact logically addresses the question of the possibility to see breakthrough changes in the near future.

4. Membrane gas separations tomorrow

4.1. Overall framework

Coming back to the conditions that historically enabled membrane applications to become a reliable solution, three important prerequisites, among others, can be identified.

- Access to selective and permeable materials
- Ease of materials processing and availability of fiber and module production technologies
- Robust process methodology (in order to correctly evaluate the best cost-effective design and associated operating conditions)

On the other hand, the selection of the most appropriate industrial problem that matches the set of existing separation performances possibilities is also of utmost importance. The industrial needs/technological solution interplay has thus to be continuously explored, in order to select a possible new interesting application. This simplified framework is sketched on Fig. 1.

It is obvious that many other conditions, including political, historical or economical context, contribute to the application of a new technology [12]. The story of the long road from invention to industrial use is beyond the scope of this very focused perspective. It is however interesting to analyze the recent changes on the three domains shown on the left side of Fig. 1.

4.2. New materials: the nanostructured membrane revolution

The key performance indicators (KPI) of a membrane material candidate are quite simple and logical: a high enough separation potential is needed, classically expressed for a binary mixture as the ratio of the permeabilities of the two compounds. The so-called membrane material selectivity α is obtained and will play a key role in the process separation performances possibilities:

$$\alpha = \frac{P_i}{P_j} = \frac{S_i D_i}{S_j D_j} \quad (1)$$

A second KPI corresponds to the production capacity of the material. For a given compound/material pair, the pure gas steady state flux J_i that can be obtained for a given surface area and driving force (Δp_i) is used for that purpose (equation (2)). The key material indicator is then the membrane permeance for compound i L_i .

$$J_i = L_i \cdot \Delta p_i \quad (2)$$

For a thick selfstanding material, or a negligible support resistance, the permeance of the material L_i can be related to the intrinsic active layer permeability P_i (most often expressed in Barrer, 1 Barrer = 10^{-10} cm³ STP. cm⁻².cm. s⁻¹. cm Hg⁻¹) and effective layer thickness (z) through:

$$L_i = \frac{P_i}{z} \quad (3)$$

For gas separations, a somehow weird permeance unit, namely Gas Permeation Unit (GPU), is used: 1 GPU = 10^{-6} cm³ STP. cm⁻². s⁻¹. cm Hg⁻¹. With this set of units, one GPU corresponds to an active layer of 1 μ m with a material showing a permeability of 1 Barrer, with a negligible support resistance. Nevertheless, the impact of support resistance is most often significant for materials used in industrial applications. In that event, an experimental measurement of L is more appropriate than the above expression. Early commercial membranes showed experimental permeance data for gases down to a few GPU's, while recent data above 2000 GPU have been reported for CO₂ with commercially available membranes (such as Polaris from MTR [19] or Polyactive from Hereon [20]).

The membrane material quest for gas separations has thus been mostly achieved around the search for maximal selectivity and permeance. Nevertheless, it appears from experimental data sets that these two characteristics are antagonistic, at least for dense polymers. A so-called trade-off curve, originally suggested by Robeson [21], has to be taken into account and it is specific of a given gas pair for polymeric materials. A trade-off curve prediction possibility is offered based on correlations for gas solubilities and diffusion coefficients in polymers [22]. For a specific polymer however, even though significant progress has been made in the prediction of gas permeability through correlations or molecular simulations approaches, the absolute need of precise and effective permeance data for reliable process simulations still requires the use of experimental methods. A very large ratio of reported permeability experimental data corresponds to pure gas data obtained on thick, homogeneous membrane samples. A relevant process simulation requires however effective experimental data obtained on a real membrane (i.e. with a sample combining support and thin layer), for a gas mixture, ideally under the temperature and pressure conditions used for simulation. Very few data fulfilling these conditions are available in the open literature.

Trying to overpass the selectivity/permeance trade-off boundary of dense polymers is the main target of membrane material researchers for decades. A series of strategies have been tested: new structures such as Polymers of Intrinsic Microporosity (PIM) [23], new preparation techniques such as Thermally rearranged Polymers (TRP) [24] or hybrid

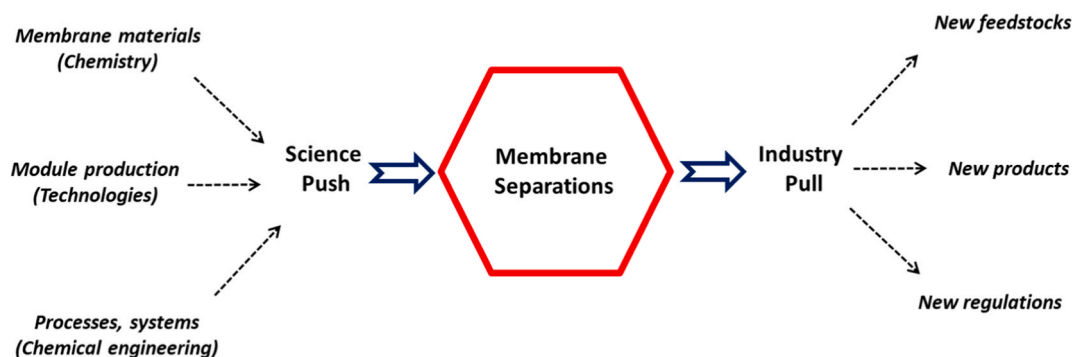


Fig. 1. Illustrative representation of the different drivers leading to membrane gas separation developments. New scientific methods (left) can produce novel materials, manufacturing technologies or process design tools. New resources or marketed products or regulations generate emerging needs from industry (right). Membrane gas separation development requires the materials science and process engineering inputs to be efficiently associated in order to identify the best (i.e. the most cost-effective) solution.

materials including a dispersion of particles in a polymer phase, such as Mixed Matrix membranes (MMM) [25]. Organic-based nanofillers, such as macrocyclic cavitands, have also demonstrated strong potential for enhancing gas separation performance [26]. Nevertheless, the most impressive improvements, be it for selectivity or permeance, have been obtained by nanostructured inorganic materials such as zeolites [27], carbon nanotubes (CNT) [28], carbon molecular sieves (CMS) [29], graphenes (GO) [30–32] or metal oxide frameworks (MOF) [33], among others. For high temperature applications, perovskite membranes for oxygen separation [34], metallic membranes (Ni and Pd) for hydrogen separation [35], and molten salt membranes for CO₂ permeation [36] are also intensively investigated with promising results. A detailed analysis of the thousands of data reported for different gases and target separations is beyond the scope of this paper. It has to be highlighted however that a large set of materials, most often at lab scale, offered improvements of one to two orders of magnitude in permeance or selectivity compared to polymers (Fig. 2).

A different strategy, explored mostly by petroleum companies a long time ago, consists to make use of a chemical reaction in order to increase selectivity and permeance. Experiments and modelling studies on chemically reactive membranes can be dated back to the 50's. Stability and poisoning issues have very often blocked their practical use in industry [4]. Basically, two approaches can be tested: a chemically reacting liquid dissolved or supported in a compartment (Liquid Membranes LM), or grafting a chemically reacting moiety on a polymer backbone (Fixed Site Carrier Membrane FSCM). A few companies currently explore novel materials and modules for carbon capture with FSCM [38], liquid membranes [39] or hydrocarbon separations [40].

Transforming a novel material into an effective membrane (thin layer on a support) from lab scale to industrial level is often seen as a death valley. This statement applies to polymers but also for inorganic materials. Generally speaking, nanostructured inorganic materials indeed address formidable challenges related to scalability and cost efficiency that greatly limit their widespread application today. It has to be noted however that several companies propose today membrane modules based on high performance inorganic materials [41–44].

Coming back to Fig. 1, it is clear that a new set of membrane materials showing breakthrough improvements in terms of selectivity or

permeance now exist for gas separation applications, with a very limited number of commercially available inorganic materials. It is likely that inorganic membranes will more significantly enter the field soon or late, in a context quite similar to what happened for water treatment some decades ago (competition between cheap, easy to process polymers and costly, high performance inorganic membranes).

From an engineering point of view, increasing material selectivity or permeance is undoubtedly of key interest for gas separations, but generates important changes.

First, a high selectivity (y axis on Fig. 2) will translate into a larger surface area because the fast compound driving force quickly vanishes [9]. This somehow counterintuitive effect was early identified especially for drying operations and requires a driving force carrier gas technique or partial retentate recycle loop in order to prevent rapid flux decrease [45].

Second, a high active layer permeance (x axis on Fig. 2) will logically decrease the active layer intrinsic mass transfer resistance. Above a certain level, two complications arise. On the downstream part (permeate) of the active layer, the support resistance plays an increasing role and a threshold overall permeance, which can be dominated by the support mass transfer coefficient, will result [46]. Mass transfer limitations within the substrate is often closely linked to intrusion, which can significantly hinder overall performance. Strategies to mitigate substrate resistance through morphology adjustment, such as modulating surface hydrophilicity and coatability for composite hollow fiber membranes are thus of major interest for reducing support resistance [47]. Besides the support resistance issue, another problem occurs on the upper part (retentate) of the active layer for high permeance membranes: a gas phase boundary layer resistance, so called concentration polarization, impacts the effective transmembrane flux. This phenomenon is well known in membrane science and is almost systematic for liquid separations [48]. For gas separations, it has been reported to be significant in some cases [49,50] but the process design methodology very often postulates the membrane to be the only mass transfer resistance [9,10]. Fig. 3 shows the mass transfer coefficient ranges for gases in free flow, gases in porous media, liquids in porous media [51] compared to the very large hypothetical membrane active layer intrinsic coefficient (i.e. active layer permeance). The 1000 GPU limit and 100 selectivity limit has been proposed for polarization concentration to be taken into account [52]. CFD simulations show that the situation is much more complex and the interplay between permeance and selectivity levels can

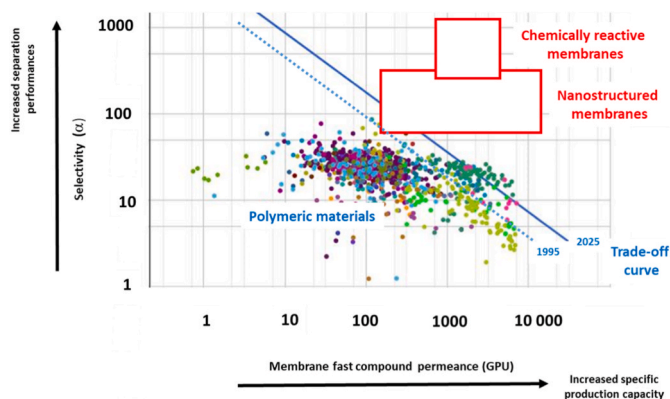


Fig. 2. Sketch (not at scale) of the qualitative evolution of membrane gas separation materials for a gas pair separation (from Ref. [37], adapted). The trade-off limit, specific of polymers, gradually shifted over the years thanks to advanced approaches. Non polymeric nanostructured materials, chemically reactive membranes can overpass the trade-off curve with breakthrough permeance or selectivity. A very large part of available data and reported trade-off curves are based on pure gases permeabilities obtained on thick selfstanding materials. The translation in effective permeance (x axis), potentially obtained with thin layer supported membranes is subject to caution. Additionally, major changes in the performances, especially selectivity, often occur with mixed gas conditions.

(1 GPU = 10^{-6} cm³ STP. cm⁻². s⁻¹. cm Hg⁻¹).

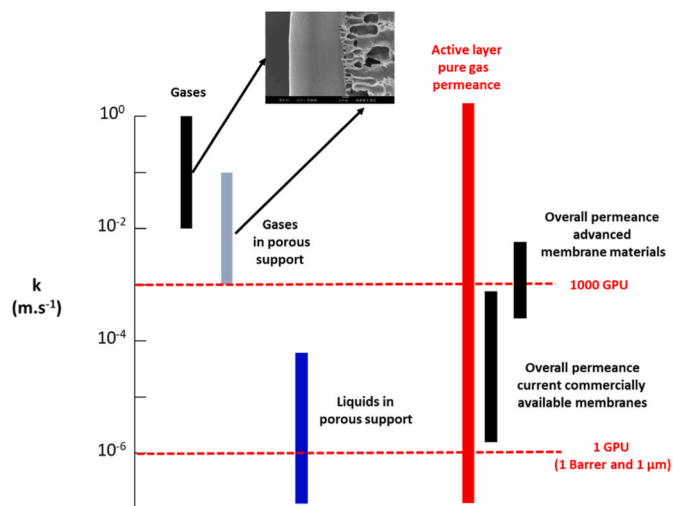


Fig. 3. Comparison of the range of mass transfer coefficients in gases (retentate side), gases in porous media (support resistance), liquids in porous media (in case of capillary condensation of heavy compounds in porous support) and for the active layer material (with the corresponding GPU level). (1 GPU = 10^{-6} cm³ STP. cm⁻². s⁻¹. cm Hg⁻¹).

have a significant impact on separation performances compared to the 1D approach even for lower values [53].

Fig. 3 shows that the 1D negligible concentration polarization simplifying hypothesis is likely to hold for most of the commercially available membranes (except for a very high permeance level). The risk associated to capillary condensation effects in the porous support is noticeable. For the highly permeable materials discussed above, there are two issues to be solved in order to express the potentiality of the active layer: a very permeable support (making use for instance of new membrane preparation methods [54,55] or, if possible, a thin self-standing active layer) and a maximal retentate side mass transfer coefficient. Moreover, for highly selective materials, the driving force challenge sketched above requires dedicated solutions. These challenges could be partly taken into account through novel module designs, discussed in the next section.

4.3. Module manufacturing

Besides novel materials design, the second driver shown on Fig. 1 points out the portfolio of material fabrication techniques, which are available for module production. The membrane gas separation modules existing today are extremely sophisticated objects, requiring a high level expertise and know-how in materials processing, thin layer technologies, cutting and gluing operations. This explains why very few companies exist today as equipment suppliers in the field. A typical gas separation module is described on Fig. 4, which shows microfluidic type dimensions for gas flow and nanoscale range for the active layer. In terms of design, a 1D axial flow and a constant cross section area geometry is used for process operation, with co or counter-current flow possibility, similarly to tubular heat exchanger technology. The impressive specific surface area (up to 20 000 m² per m³) offers volume reduction opportunities that are explored for process intensification purpose. This area per volume specificity, which can be achieved only with polymers today, is of major interest when minimal volume and/or weight is needed (such as for space, off shore or embarked systems).

Because the mass transfer resistance of the process is located in the membrane (+support), module process design priority is maximizing the specific surface area with simple fluid flow solutions. With the advent of high performance/high selectivity materials, the increasing resistance of the gas boundary layer calls for revisited module design constraints. The liquid membrane module development is faced to this situation from the beginning [4]. Somehow, the gas membrane module requires now to include similar constraints.

Turbulence promoters through tailor made spacers in the retentate gas compartment are one possible solution, which is applied for a long time for liquid membrane separations. Membrane surface modifications with *in situ* turbulence promoters such as corrugated or striated surfaces (in place of the smooth active layer shown on Fig. 3) can also be of interest. Helical shape fibers may also offer attracting performances, combining increased mass transfer coefficients and high specific surface area (Fig. 5a) [56]. Alternatively, a completely different membrane and module geometry could also be attempted, in order to generate fluid redistribution conditions. Unfortunately, coating and extrusion

techniques limit possibilities today mostly to sheets and cylinders. With the advent of 3D printing technologies, breakthrough changes could occur. Impressive developments have been reported [57–60], possibly opening the way to thin skin membranes, direct fiber production up to integral 3D printing module production (Fig. 5b).

From a more exploratory point of view, it is interesting to note that mass exchangers in living systems (e.g. kidneys, lungs, leaves, *rete mirabile*) always make use of complex geometries. Different scale levels, such as constructal flow systems (Fig. 5c), different cross section areas, and sophisticated fluid flow redistribution are typical of biological mass exchangers (Fig. 5d). Bioinspired gas separation module design studies are however essentially unexplored, due to the easier construction achievable with flat or hollow fiber geometries.

Important efforts are needed in order to face the module development approaches sketched above. Experimental challenges combined to computational studies (ideally making use of CFD) will be necessary to evaluate the module performance enhancement possibilities and production limitations. Heat exchanger developments show the interest of fluid redistribution systems (microstructured or chaotic) in order to increase transfer performances when the resistance is located in the fluid boundary layer. Significant improvements are reported with constructal [61,62] or lung inspired [63,64] heat exchangers. As discussed above, a similar boundary layer resistance location holds for high performance membrane gas separation modules. There is for sure a long way before the interest, or not, of these prospective developments to be analyzed, and the novel geometry research objects to possibly become a liable industrial solution. The prospective benefits of these exploratory approaches clearly remain to be proven.

4.4. Process design strategy and methodology

It is generally stated that a good process can save a poor membrane, in order to highlight the absolute necessity to apply a rigorous and efficient design methodology to get the best set of membrane material/flowsheet/operating conditions that reaches the separation specifications.

The generic target of process engineering studies is to identify the best design (i.e. flowsheet) and set of operating conditions for a given set of specifications and feed conditions. Simulation tools have been proposed for a long time in order to achieve that purpose. The key computation operation is the rigorous local partial flux (J_i) calculation. The permeability (P)/permeance (L) concept, initially proposed by Graham, is the backbone of simulators:

$$J_i = L_i \cdot \Delta p_i = L_i \cdot (x_i p' - y_i p'') \quad (4)$$

Where L_i stands for the effective permeance of compound i , p' for upstream pressure, p'' for downstream pressure, x_i for upstream mole fraction of the component i , y_i for downstream mole fraction of the component i . For a 1D upstream and downstream side flow system, the local partial flux term expression can be coupled to mass balances in order to evaluate the axial retentate and permeate compositions. Different flow conditions can be postulated such as cross plug flow, co-

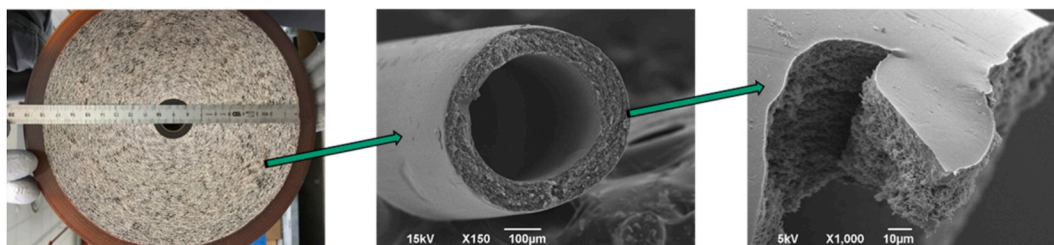


Fig. 4. Example of an industrial gas separation module containing thousands of hollow fibers (left), with internal diameter around 300 μm (middle) and dense skin active layer around 20 nm thickness (right).

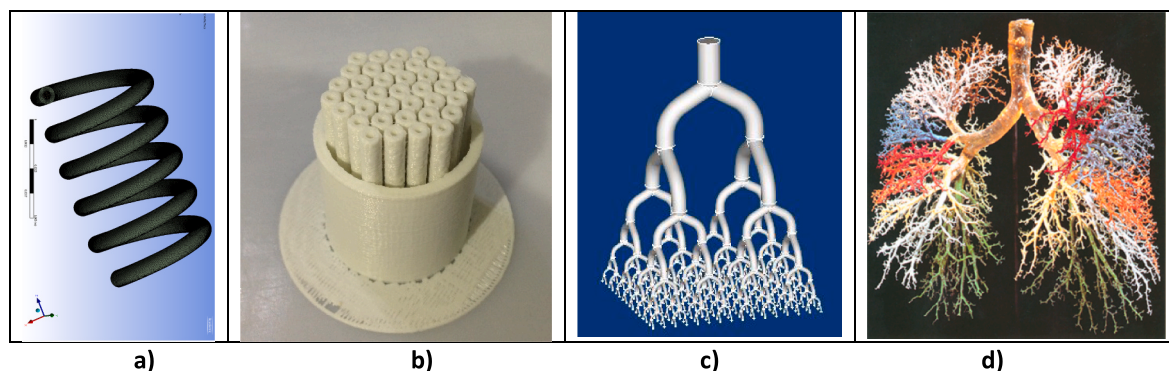


Fig. 5. Exploratory membrane module development concepts: a) helical fibers, b) 3D printed self-standing hollow fiber module, c) constructal exchanger, d) microstructured fluid flow gas system (lung example).

current plug flow or counter-current plug flow and the calculations extension to multicomponent separations problems is straightforward. Weller & Steiner first formulated the set of equations [65] that still serve as mathematical framework for process design computations [66,67]. A very good prediction of experimental results is generally obtained, providing the right input, namely the effective permeance of each compound L is used. A variable permeance behavior (i.e. permeance variation as a function of partial pressure conditions) complicates the simulation approach, but rigorous solutions exist [68]. The most difficult complication happens when flux coupling (i.e. influence of one compound flux on the other compound flux) occurs. No generic computation strategy is available in that case in order to predict the coupling coefficients.

It is important to note that, besides the downhill transport direction indicated in equation (4), recent studies report uphill transport mechanisms, in which one species moves against its concentration gradient through coupling with the downhill transport of another [36]. This new type of separation concept opens promising perspectives that remain to be explored through rigorous process design approaches.

The early attempts to achieve process design calculations in the 60's made use of approximate analytical solutions to the set of algebraic-differential equations. Tailor made computer codes extended the possibilities in the 70's and today membrane gas separation systems can be easily simulated thanks to process simulators [67]. The identification of the most efficient design for a set of specifications is a classical problem and the number of stages is very limited. While equilibrium based processes can operate with tens (liquid extraction), hundreds (absorption),

up to thousands (adsorption) stages, any supplementary membrane stage requires in fact one more compressor or vacuum pump. As a consequence, membrane process flowsheets are quite simple and the great majority of industrial systems are limited to one or two stages. A maximum of three stages is used for biogas upgrading for instance Ref. [69]. Despite the limited number of stages, the connection possibilities (partial recycling loops) and range of operating conditions (variable pressure ratios) can lead to time consuming optimization studies through classical PSE (Process Systems Engineering) softwares (Fig. 5).

The trend today is to make use of process synthesis computer codes, taking advantage of advanced optimization methods [70–72]. In that case, the objective function is the overall cost of the process (including CAPEX and OPEX). Numerous studies and methods have been recently reported in this field, offering promising perspectives for cost efficient membrane solutions (Fig. 6). The impact of membrane material and cost can also be investigated thanks to these tools. For example, for NEA production, the market number one of membrane gas separations, process synthesis studies rediscovered the number of stages and flowsheets proposed by engineers for a long time as a function of target nitrogen purity [73]. Alternatively, new outcomes are obtained: vacuum pumping is shown to improve the process and, surprisingly, there is no interest to combine different membranes in multistage systems for polymeric membranes with trade-off limit performances. The ideal polymeric membrane, with selectivity permeability trade-off limitations, should be more permeable, even at the expense of a slight selectivity decrease, and should be used for all the stages.

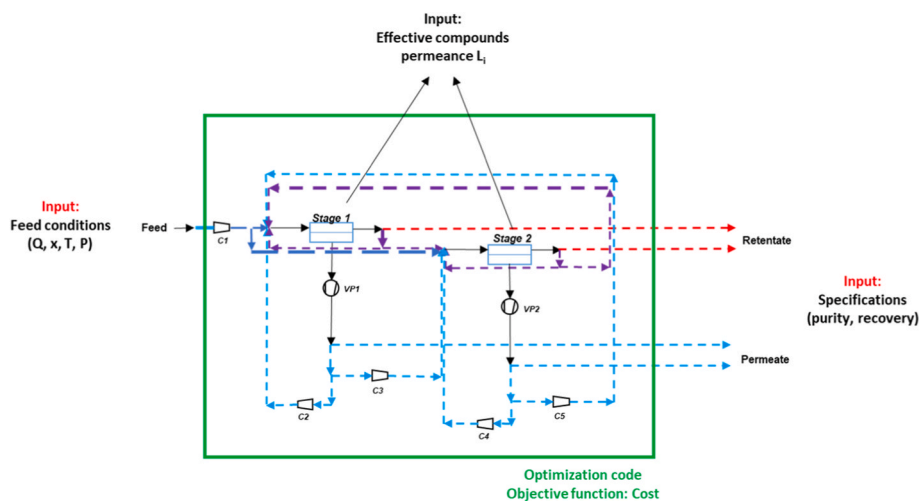


Fig. 6. Example of a two stages membrane flow sheet superstructure, including separate upstream and downstream pressure level for each stage (i.e. flexible pressure ratio) and partial recycling loops.

A very different result of process synthesis studies is obtained for biogas upgrading, one of the current markets showing the greatest increase. Classical two stages designs, with two different recycling options, are again rediscovered, depending on the selectivity of the material [74]. But, in that case, an increased selectivity is of major interest. Moreover, a high permeance, high selectivity, high price (2000 \$/m²) zeolite membrane is shown to offer a significant biomethane cost decrease compared to existing polymeric materials (polyimide, 50 \$/m²). The possibility to now quickly evaluate the impact of a new membrane for a given application, that is bridging the gap between material science and process engineering, is very promising.

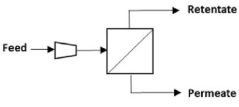
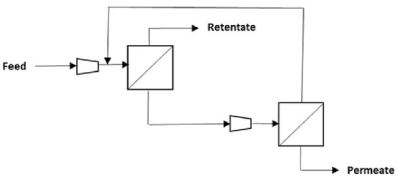
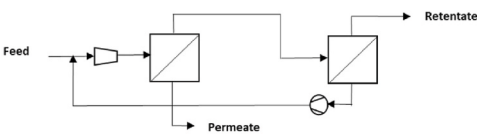
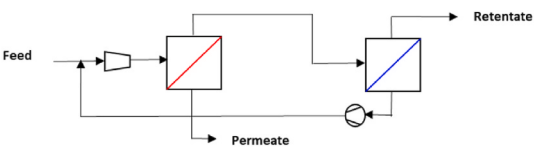
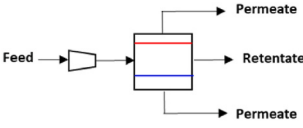
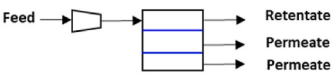
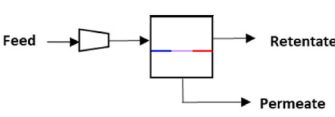
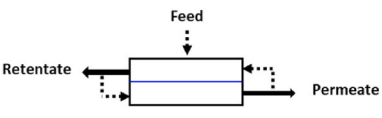
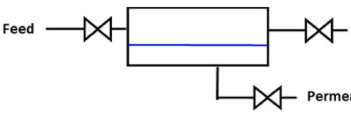
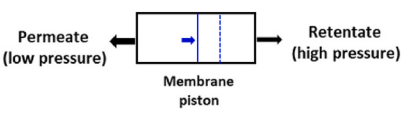
The membrane process designs operated today in industry are rather simple, as stated above. One or two stages (with one recycling loop), using the same membrane from the same supplier for a given

installation, with one compressor or vacuum pump. A material showing constant properties all over the module axial direction is used; it is interesting to note that membrane materials showing a gradual axial porosity are proposed today for membrane liquid separations. This peculiarity enables a better permeate flux equipartition, leading to a higher module productivity. A similar concept could be explored for gas separations.

More generally, the question of the potential interest of more complex designs or exploratory concepts for membrane gas separations has been investigated for a very long time. A detailed analysis of the numerous alternatives and their advantages and drawbacks is beyond the scope of this work, but the current designs and some proposed but largely unexplored concepts are summarized in Table 3.

It should be noted that, despite significant past work devoted to a

Table 3
Examples of current and exploratory membrane process designs and concepts.

Current Process Design	Single stage		[4,9]
	Two stages retentate recycle		[4,9]
	Two stages permeate recycle		[4,9]
Exploratory Design Concepts	Multistage multimembrane		[75]
	Two membranes permeation module		[75]
	Internally staged module		[76]
	Gradual axial performances membrane		[77]
Exploratory Process Concepts	Membrane column		[78,79]
	Unsteady membrane gas separations		[80]
	Membrane piston		[81]

variety of internally staged module, unsteady operation and membrane column designs for many years, no industrial application emerged and these process options are today abandoned.

Generally speaking, the identification of the best design for a given gas separation application remains an open question. Comparative studies and systematic optimization methods applied to different process options are lacking. It might be expected that, through AI methods, this complex task will be achievable in the near future.

Besides the process design challenge, driving force selection is an important aspect. This point is, to a large extent, neglected up to day. Feed compression is most often chosen for that purpose (Table 4). Two exceptions can be noted: compressed feed situations, such as found for natural gas or hydrogen recovery, which do not require a costly compressor apparatus, and vacuum pumping which is chosen in case of explosion hazards, for VOC recovery application for instance. Feed compression is interesting because it requires the lowest membrane surface area, while vacuum pumping is needed in order to minimize energy requirement [82]. This explains why vacuum is proposed for membrane carbon capture processes, because of the strong energy requirement constraint [83].

Generally speaking, different driving forces can be proposed for membrane gas separation besides compression or vacuum [9]. Power input (i.e. electricity) clearly is a very efficient process solution but it is considered as a drawback in terms of energy requirement compared to thermal separations [13]. A second law efficiency comparison indeed typically makes use of a 2.7 conversion factor between thermal energy and power [82]. Moreover, compressors are costly and often correspond to the dominant contribution of the overall cost of membrane processes [4,9]. In place of compression or vacuum pumping, a permeate dilution effect can be used, shown on Fig. 7, similarly to dialysis operation for liquids [4]. This option can be cheaper (i.e. a fan is used in place of a compressor) at the expense of lower fluxes and possible complications such as sweep gas counter diffusion or increased mass transfer resistance, due to carrier gas accumulation in the membrane porous support. A different strategy, making use of a gas mixture sweep on the permeate side of a module, so called gas/gas membrane contactors has been recently proposed [84]. A very special strategy consisting of recycling part of the retentate to the permeate side is occasionally applied for very selective materials, such as drying operation [45]. In some cases, a diluting effect, analogous to an internal sweep can be obtained, when the feed mixture contains an unwanted, highly permeable, easy to condense compound. The role of water vapor in flue gases for post-combustion carbon capture processes is a typical example [85]. With the advent of temperature stable inorganic materials, steam sweeping operations could be of interest but remains largely unexplored.

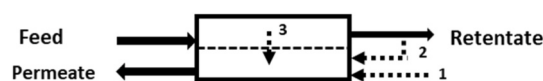


Fig. 7. Sweep operation strategies for membrane gas separations: 1. External sweep (carrier gas or gas mixture), 2. Retentate to permeate partial recycling (e.g. drying operation), 3. Internal sweep (e.g. water vapor in carbon capture).

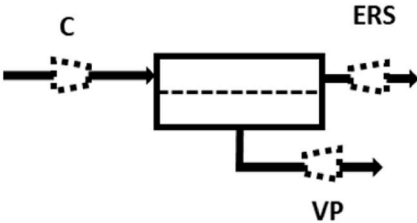
Generally speaking, sweep operation is not applied for gas separations, except the drying application. It might be however that this will be reconsidered, given the very high selectivity of advanced materials, together with increasing constraints on energy requirement. Some simulators offer now the sweep operation option [86] for process design studies. It is expected that process synthesis programs will soon include this possibility into combinatorial optimization softwares.

More exploratory driving forces are, in principle, applicable, through the generalized electrochemical potential expression used to derive the different possibilities [87]. A transmembrane temperature difference is possible, but does not generate significant transmembrane fluxes because of the tiny impact of thermo-osmosis effect explored a long time ago [88]. Electrical or magnetic fields are also expected to induce very minor effects for gas transport in membrane materials. Photochemical systems are also of interest for specific applications. For instance, a photodriven membrane process has been recently reported for diluted VOC removal in indoor pollution treatment [89]. This counterintuitive application of membranes, which are logically discarded for diluted feeds due to too low driving forces, is interesting. The key role of the membrane is here to provide a split flow; the low module stage cut produces a VOC enriched permeate flux one order of magnitude lower than the feed flowrate. This allows a significant increase of residence time in the catalysis layer, located at the permeate side of the membrane, thus improving conversion (i.e. pollutant destruction). In place of a low energy efficiency vacuum pumping, a diode light emission is applied and the associated destruction of pollutants generates the transmembrane driving force. The overall energy efficiency of this integrated system is very high. A supplementary advantage is that the membrane can play a barrier role to prevent secondary species to be mixed with the retentate gas flux. The system thus offers a steady state bulk removal of indoor pollutants, without the bed saturation and water negative capacity impact observed with adsorption cartridges. Secondary products emission is strongly limited compared to classical photocatalytic systems.

Membrane processes are usually considered to find use in the production of high purity products only when their separation factor is sufficiently large to permit one-stage separation or when they are used in conjunction with other separation technology capable of completing the task [90]. Hybrid processes including one membrane step is an

Table 4

Schematic driving force strategies for the current membrane gas applications. C: compressor. VP: vacuum pump. ERS: Energy Recovery System (expander).

Application				
	C	VP	ERS	
H ₂ recovery	–	–	–	Compressed feed
Natural gas	–	–	–	Compressed feed
Air separation (NEA)	+	–	–	
Biogas	+	–	–	Vacuum risky due to air leaks
VOC recovery	+	+	–	Limited compression due to explosion hazard
Carbon capture (post-combustion)	+	+	–	Vacuum required for energy efficiency

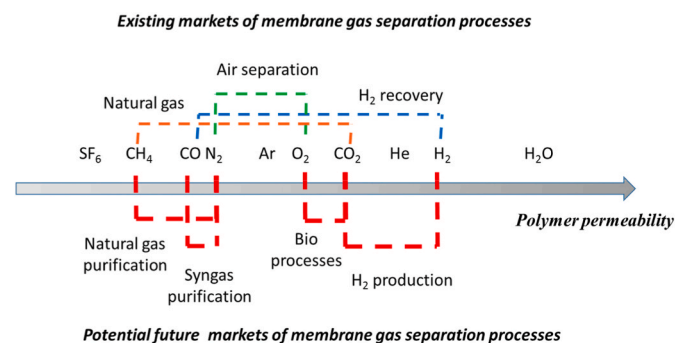


Fig. 8. Gas pairs (dotted lines) of existing and potential future markets for membrane gas separations as a function of polymer permeability ranking (rapid gases on the right scale). The future markets (down dotted lines) globally correspond to more difficult separations, thus requiring materials with improved molecular sieving properties.

Table 5

Overview of possible changes for future membrane gas applications, based on the prospective attempted analysis.

	Today	Toworror?
Materials	Dense polymers	Inorganic Chemically reactive
Modules	1D, monomaterial for active layer	Multi active layer materials Complex shape
System	Compression 1 or 2 single-membrane stages (maximum 3)	Alternative driving forces (sweep, heat, light), integrated systems Multimembrane systems
Applications	Major 5 (Table 2) Limited temperature	Carbon capture, biogas, biohydrogen, sustainable processes, biorefineries Low to high temperature

active field of research, that needs to be reevaluated with high performance materials. A membrane preconcentration function is often very interesting and very energy efficient [91]. This can be of major interest for an absorption post-treatment in natural gas processing [92], or for carbon capture by cryogeny [93]. Hybrid processes including a membrane stage should ideally be studied thanks to process synthesis methods but the joined analysis of different separation processes is complex and requires further development.

4.5. New applications prospects

The landscape of Chemical Process Industries (CPI) is changing. New feedstocks are increasingly used, renewables in place of fossil, with a possible switch from oil refinery to biorefineries in the future. New set of constraints have to be taken into account for process design, with carbon footprint, water footprint and environmental impact playing a key role. Energy use via renewables feedstocks such as biomass for syngas production, biogas or algae is promoted. Hydrogen and electricity are considered as the energy vectors of the future. These changes generate separation problems, illustrated on Fig. 8, that could be included in the portfolio of major membrane gas separation markets. The interplay between high performance materials and optimal process design for new

set of mixtures (CO/N_2 , CH_4/N_2 , O_2/CO_2 , H_2/CO_2) is currently investigated.

Several applications are expected to emerge within this new industrial framework. For instance, membranes are promising for carbon-neutral shipping (low CO_2 contents generated from marine engines integrated with onboard CO_2 liquefaction and storage for port offloading) or biogas upgrading in high-nitrogen-content streams (zeolite membranes for N_2/CH_4 separation).

Besides the gas applications, numerous vapor separation problems exist, especially for biorefineries [94]. The recovery of organic acids (acetic, butyric), hydrocarbons (acetone, isobutene, ethylene), alcohols (ethanol, butanol) from bioreactors exhaust gases, biohydrogen production are new challenges for which membranes could provide relevant solutions. Hybrid processes combining thermal processes (distillation, cryogeny) with membranes units is likely to be relevant for these applications [95].

Nevertheless, the hydrocarbon separation problems encountered in oil refineries (ethylene/ethane, propylene/propane, aliphatics/aromatics ...), with impressive energy efficiencies improvements if robust high performance membrane materials are available [18], is expected to remain of major interest for a long time.

5. Synopsis and conclusion

Membrane gas separation is a rapidly evolving field with significant potential for industrial and environmental applications. The recent advances in high performance materials, module production technologies and process integration solutions through digitalized tools are expected to drive the future.

Table 5 summarizes the possible evolutions or breakthrough developments that could occur in the near future, based on the overview proposed above. This list of expected changes is obviously highly prospective. A key issue remains indeed related to durability, scalability, and sustainability, which are crucial for the widespread adoption of membrane gas separation technologies. Nevertheless, looking back the breakthrough innovations performed since 50 years, Danckwerts quotation is probably relevant for the forthcoming decades: "It will be a great mistake to think of the content of chemical engineering science as permanently fixed. It is likely to alter greatly over the years in response to the changing requirements of industry and to new scientific discoveries and ideas for their application" [96].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors gratefully thank the 'French ministry of higher education and research' and 'Carnot ICEEL' (CREA-TIX innovation program) for support given to this study.

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List of symbols

D_i	Diffusion coefficient ($\text{m}^2.\text{s}^{-1}$)
J_i	Transmembrane flux ($\text{mol}.\text{m}^{-2}.\text{s}^{-1}$)
L_i	Permeance ($\text{mol}.\text{m}^{-2}.\text{s}^{-1}.\text{Pa}^{-1}$)
P_i	Permeability of the component i ($\text{mol}.\text{m}^{-2}.\text{m}.\text{s}^{-1}.\text{Pa}^{-1}$)
p'	Total upstream pressure (Pa)
p''	Total downstream pressure (Pa)

S_i	Sorption (solubility) coefficient ($\text{mol}\cdot\text{m}^{-3}\cdot\text{Pa}^{-1}$)
x_i	Mole fraction of component i in the retentate flow (–)
y_i	Mole fraction of component i in the permeate flow (–)
z	Active layer thickness in the membrane (m)
α	Membrane selectivity (–)

Data availability

No data was used for the research described in the article.

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